

P K G - X X X

Discovery Maturity Assessment

— Cycle 69

Engineering Concept Package — produced per MASTER_PROTOCOL.md

PACKAGE ID

PKG-XXX

PACKAGE MATURITY

EVALUATION

DATE

2026-08-03

GOVERNANCE

MASTER_PROTOCOL.md (11 sections)

SCANNER

Law 27/28/29 compliant

70

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Generated: 2026-08-05T03:16:52.546036+00:00 **Current phase:** I

9-Phase Maturity Scale

PHASE	NAME	STATUS	SUCCESS CRITERION
I	Scientific Memory	70%	Everything becomes replayable
II	Dimensional Reasoning	5%	Impossible laws disappear automatically
III	Symbolic Discovery	20%	Discover equations you never programmed
IV	Mechanism Induction	25%	The system explains
V	Intervention Search	30%	The engine proposes experiments
VI	Laboratory Closure	10%	The engine learns from reality
VII	Adjacent Possible Exploration	15%	The system explores what does not yet exist
VIII	Discovery Economics	0%	maximize(expected_information_gain)
IX	Apollo Benchmark	2/100 blind tests	Blind tests → 100, novel hits → 25, closed loops → 1000

Apollo Metrics

METRIC	CURRENT	TARGET
blind_tests	3	30
novel_hits	1	10
retrievals	1	10
null_results	1	10
target_blind_tests	100	100
target_novel_hits	25	25

Current Phase Detail

Phase I: Scientific Memory Status: 70% **Success criterion:** Everything becomes replayable

Assessment: IdentityGraph, SimilarityGraph, MechanismGraph, CausalGraph, ExperimentGraph exist. Missing: Observation, Intervention, Theory classes. CausalEdge has mechanism_status but no Observation with measurement/uncertainty.

Next work: Add Observation, Intervention, Theory dataclasses to invention_compiler/. Wire them into the DiscoveryGraph as new node types.

Required New Classes

- Phase I (Scientific Memory): `Observation` —  missing
- Phase I (Scientific Memory): `Intervention` —  exists
- Phase I (Scientific Memory): `Theory` —  missing
- Phase II (Dimensional Reasoning): `Dimension` —  missing
- Phase IV (Mechanism Induction): `Mechanism (expanded)` —  missing

The Final Architecture

OBSERVE → EXTRACT → REPRESENT → EXPLAIN → DISCOVER → INTERVENE
→ PREDICT → EXPERIMENT → MEASURE → LEARN → REVISE

Path Forward

The system is at Phase I (Scientific Memory, 70%). The next step is to add Observation, Intervention, and Theory dataclasses to close Phase I, then proceed to Phase II (Dimensional Reasoning) — the biggest omission identified by the CEO.